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Low-Hour, False Head Gasket Oil Leaks

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This document provides information to assist the field in correctly diagnosing low-hour, false head gasket oil leaks on high horsepower engines.

Investigation into reported oil leaks from cylinder head gaskets on high horsepower engines has revealed that the “leaks” are actually liquids used during new engine assembly that are “baking off” from heat. Analysis has shown that these assembly lubricants can come from a variety of sources:

- 140-weight oil is applied to the heads and threads of the cylinder head capscrews during engine assembly. While residual oil from piston/ring installation is wiped from the head deck of the block, it is **not** cleaned with solvent.
- Anti-seize compound is used on the threads of the exhaust manifold capscrews.
- Preservative oil is applied very heavily to the exterior of the cylinder head after assembly at the Cummins plant in Mexico.

As the engine is operated and these fluids are heated, they become less viscous and begin to flow following gravity. Engine production test does **not** provide a sufficient amount of time for the fluids to be baked off. After the engine is placed into service, the fluids are subjected to heat over a longer period and turn dark. In addition, these fluids can “bubble up” due to the heat and give the impression of being combustion leaks. The forces of gravity and fan windage tend to cause these streaks to be greater at the rear of the engine. Field experience indicates it can take 500 to 1000 hours (two to three oil changes) for all of these fluids to bake off.

If an oil leak is suspected, reference the relevant procedure in the Service Manual, for that engine model. Adding dye to the oil and performing a black light test will reveal whether the fluid is really engine oil or assembly lubricant. Oil is leaking **only** if dye is visible.

There is no value in changing the cylinder head gaskets or other gaskets if no oil leak is indicated with the black light test. Analysis of several gaskets returned from the field indicates no problem with the gasket. None have been found with a tear in the water or oil grommets that would account for the leak. The best repair strategy is to ask for the customer's patience in cleaning the streaks from the engine for the first two or three oil changes. Eventually, the bakeoff will be completed and the streaks will stop.

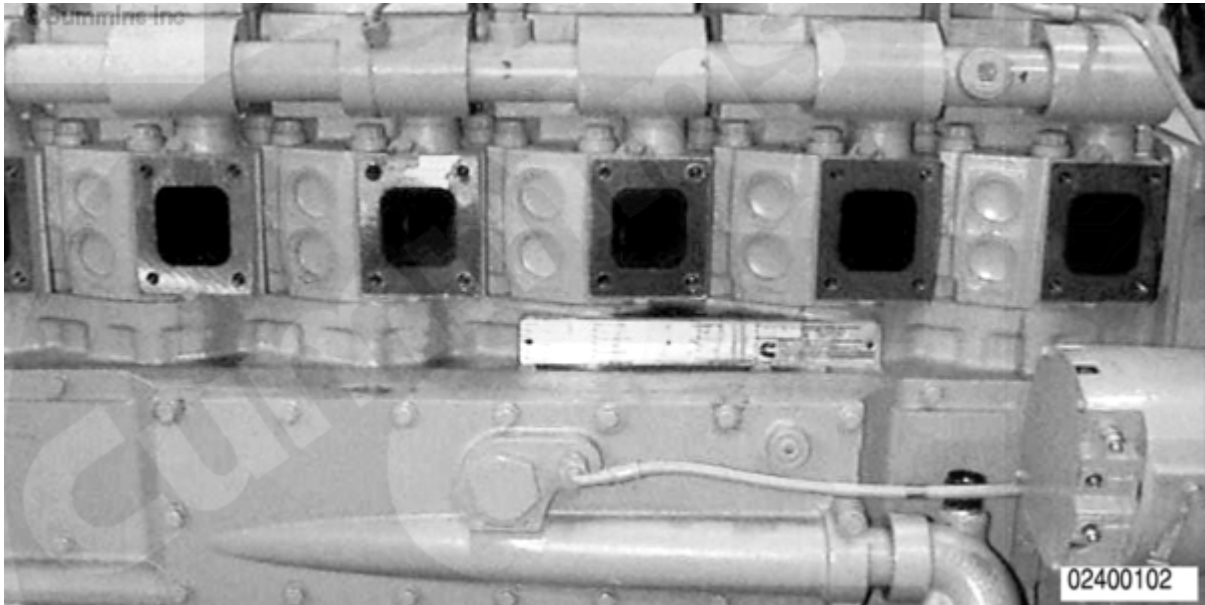


Fig. 1, Multiple Stain Paths beneath All Cylinder Head Exhaust Ports



Fig. 2, Thick Black Streaks between the Cylinder Head and Cylinder Block Deck Are Not Engine Oil



Fig. 3, Small, Thin Streaks First Appear after Engine Heats Up



Fig. 4, Dripping Near the Exhaust Manifold Where It Bolts into the Cylinder Head

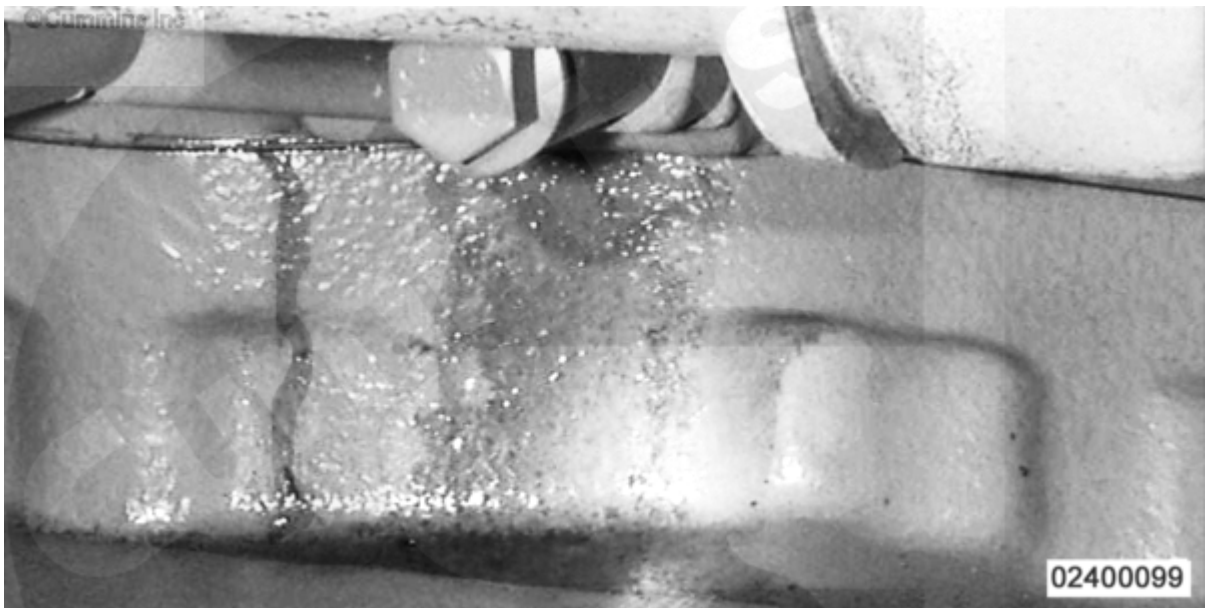


Fig. 5, Area between the Oil Cooler Housing and Exhaust Manifold Exhibits a Sheen Appearance

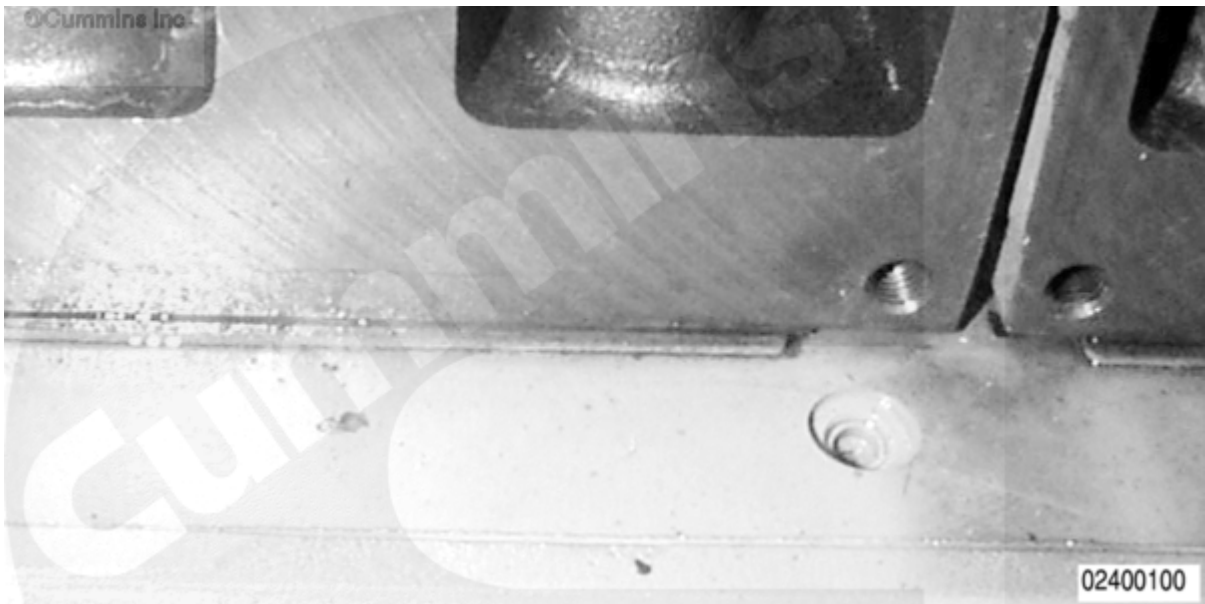


Fig. 6, Dark Brown Substance underneath Cylinder Head Intake Ports Is Not Engine Oil



Fig. 7, Multiple Stain Paths at the back of Number 6 Cylinder Head

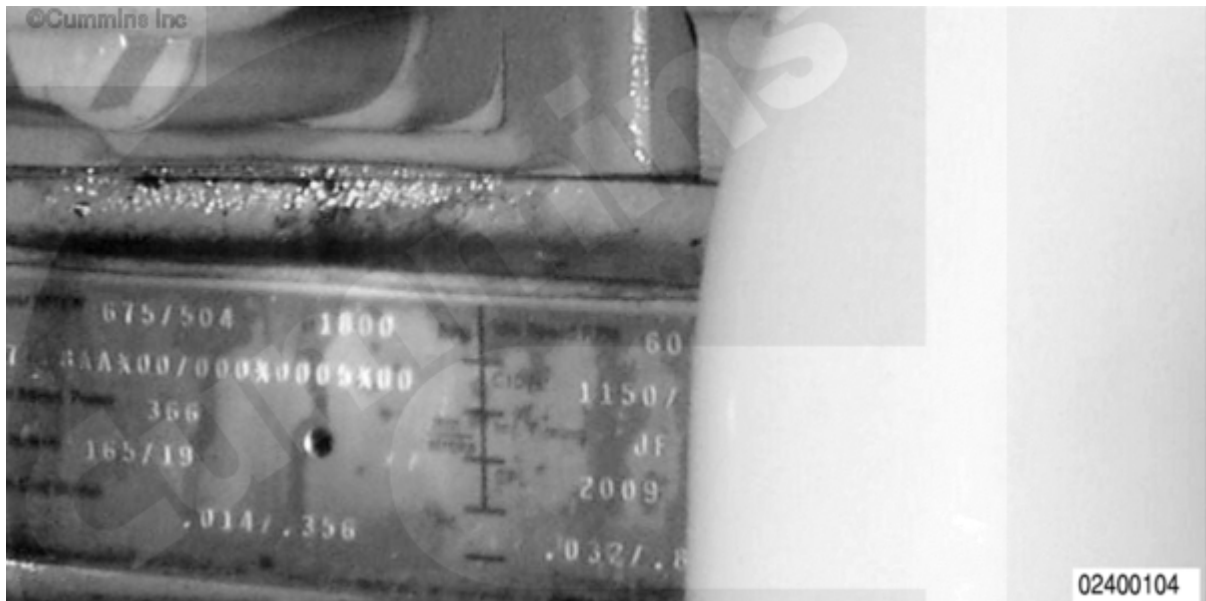


Fig. 8, Exhaust-Side Drip Stains on and around Engine Dataplate



Fig. 9, Numerous Black Streaks between the Oil Cooler and Exhaust Manifold Are Not Engine Oil Leaks

Document History

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